

Replication of Topological Data Analysis of Financial Time Series: Landscapes of Crashes

Statistics 565 - Final Presentation

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Paper: Gidea & Katz (2017), *Topological Data Analysis of Financial Time Series: Landscapes of Crashes*

Core question: Can we use topology to detect financial crises before they happen?

Why this is interesting:

- Goes beyond standard volatility/correlation diagnostics
- Uses shape-based features of data clouds, not only moments (i.e. mean, variance, etc.)
- Tests whether geometry/topology captures regime shifts before crashes

Replication Setup

Source: Yahoo Finance daily adjusted close data

- S&P 500 ($\sim$ GSPC), DJIA ($\sim$ DJI), NASDAQ ($\sim$ IXIC), Russell 2000 ($\sim$ RUT)
- VIX ($\sim$ VIX) used as benchmark comparison signal

Replication sample

- Raw prices pulled over 1987-12-23 to 2016-12-08 (pipeline-aligned)
- Effective return sample in notebook: 1992-01-03 to 2016-12-08
- Working matrix: 4D daily log returns

$$r_{i,j} = \log\left(\frac{P_{i,j}}{P_{i-1,j}}\right)$$

Object of interest: rolling *clouds* of multivariate returns, not a single series

- At each date t , take the most recent $w \in \{50, 100\}$ days of 4-index returns.
- This gives a point cloud $X_t \subset \mathbb{R}^4$: one point per day, four coordinates per point.
- We ask how the cloud's *shape* changes as markets move from calm to stress.

Why this setup is sensible

- Crises are multivariate phenomena (co-movement, dispersion, nonlinear interaction).
- Shape-based summaries can capture structure missed by mean/variance alone.
- Rolling windows let the geometry evolve through time, enabling early-warning tracking.

Step 1: Persistent homology on each cloud

- Build Vietoris–Rips complexes as radius ϵ grows.
- Track H_1 features (loops): birth/death pairs (b_k, d_k) .
- Output: persistence diagram $D_t^{(1)}$ for each date t .

Step 2: Convert to stable numeric summaries

- Transform each diagram into a persistence landscape λ_t .
- Compute $\|\lambda_t\|_1$ and $\|\lambda_t\|_2$.
- Output: two time series of topological “market stress” indicators.

Step 3: Early-warning diagnostics

- Rolling variance and low-frequency PSD of L_1 norm.
- Kendall τ trend tests on pre-crash windows.

Quick Topology Guide

Point cloud X_t

- A set of points in $\mathbb{R}^4$: each point is one trading day of 4-index returns.

Vietoris–Rips complex (scale ϵ)

- Connect points that are within distance ϵ ; this builds a network/shape proxy.
- Increase ϵ to see how connectivity and loops appear/disappear.

Persistent homology (H_1)

- Tracks loop features through scales; each has birth b and death d .
- **Birth** b : the first scale where a loop appears in the Rips complex.
- **Death** d : the scale where that loop gets filled in/merged and no longer exists.
- **Lifetime** $d - b$: persistence of that feature across scales.
- Short lifetimes are typically local noise; long lifetimes are stable global structure.
- Output is a persistence diagram (all (b, d) pairs), then landscape norms that summarize overall topological complexity over time.

Why Do This

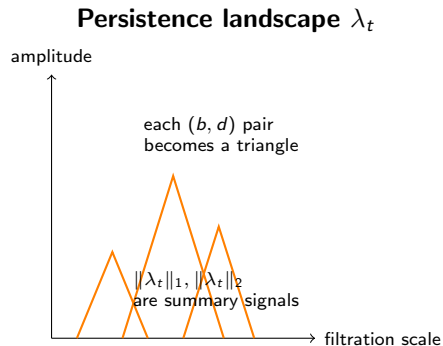
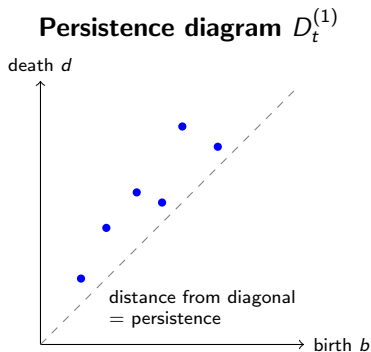
Why not just variance/correlation?

- Those are low-order summaries; they can miss changes in multivariate geometry.
- Crisis buildup can alter cloud *shape* before level/volatility fully breaks.

What the topological map gives us

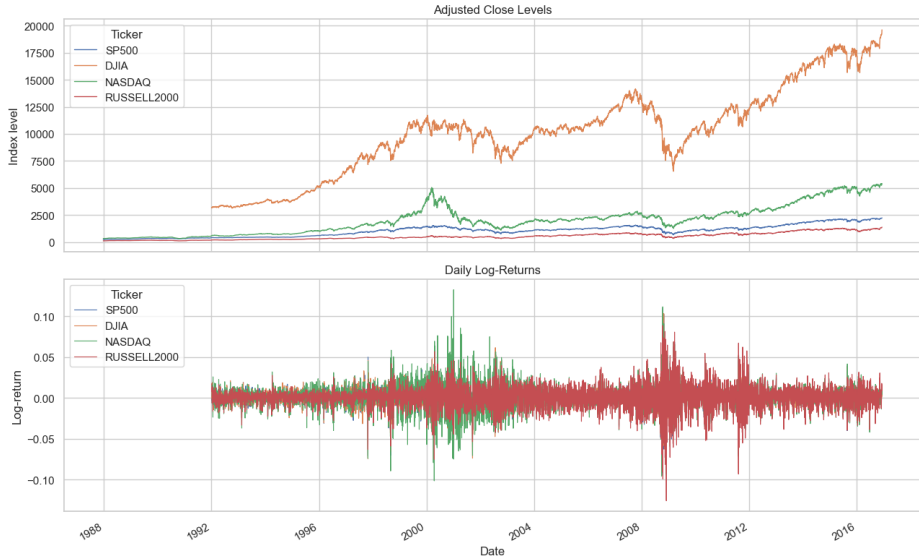
- For each date t : a persistence diagram $D_t^{(1)}$ (all loop lifetimes at once).
- A stable functional form λ_t (persistence landscape) for computation/comparison.
- Scalar indicators $\|\lambda_t\|_1, \|\lambda_t\|_2$ that become time-series risk signals.
- Testable outputs: trend statistics, event-window comparisons, and cross-signal checks vs VIX.

Persistence Diagram and Landscape (Graphic)

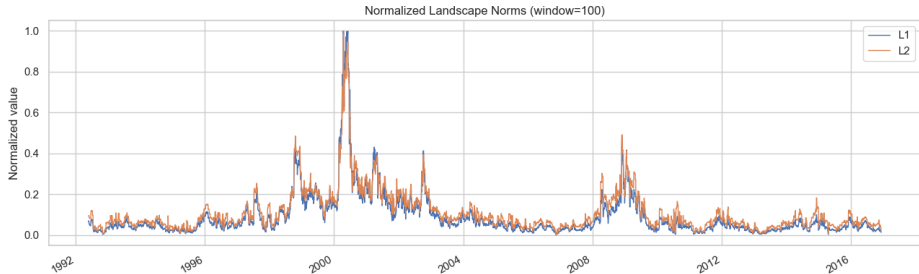
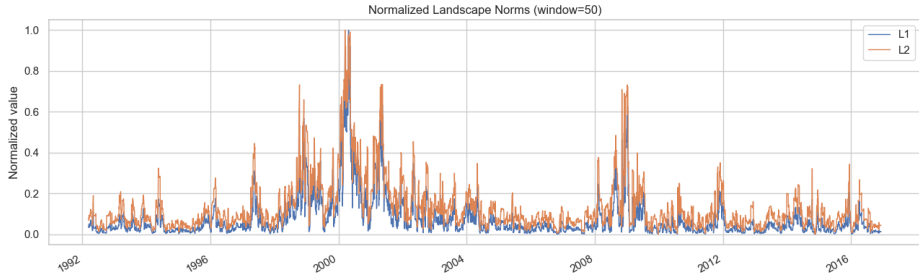


Left: raw topological features. Right: smooth representation used to compare windows and build risk-indicator time series.

Input Data Snapshot



Landscape Norm Time Series

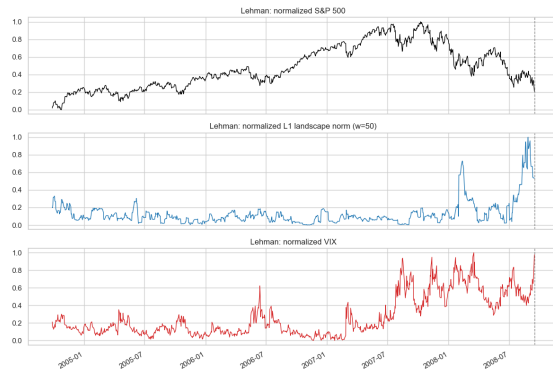
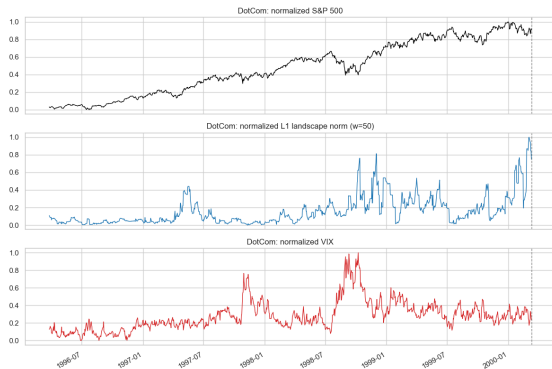


Why the Output Is Meaningful

Interpretation logic

- Calm regimes: cloud geometry is tighter/more regular $\Rightarrow$ smaller topological complexity.
- Transition-to-stress regimes: cloud deforms, dispersion rises, interactions become irregular.
- Persistent homology records this deformation through longer-lived H_1 features.
- Landscape norms compress these geometric changes into a trackable scalar signal.

Dot-Com vs Lehman



Each panel shows normalized S&P 500, normalized L_1 landscape norm ($w = 50$), and normalized VIX before event date.

Quantitative Trend Diagnostics

Event	Variance trend	Low-freq PSD trend
Dot-com	$\tau = -0.223, p = 1.44 \times 10^{-7}$	$\tau = -0.602, p = 1.42 \times 10^{-45}$
Lehman	$\tau = 0.888, p = 4.36 \times 10^{-97}$	$\tau = 0.691, p = 1.40 \times 10^{-59}$

Interpretation

- Lehman pre-window strongly supports increasing-risk narrative.
- Dot-com shows significant but opposite-direction trend in this run.
- This is an important replication nuance vs a uniform early-warning claim.

Comparison to Course Methods

Methods common in course

- Volatility modeling, correlation structure, spectral/time-series diagnostics.

What TDA adds

- Geometry of multivariate return clouds beyond second-order summaries.
- Potential sensitivity to structural regime change in cross-asset interactions.
- A complementary indicator family rather than a replacement.

Practical framing

- Best used as part of a signal ensemble with standard risk indicators.

Drawbacks and Limits

Methodological limitations

- **Parameter sensitivity:** window length, filtration scale, grid resolution.
- **Interpretability:** loop features are less intuitive than volatility/correlation.
- **Computational cost:** rolling persistent homology can be expensive.
- **Event heterogeneity:** not all crises produce the same trend direction.

Empirical limitations

- Single market (US equities) and limited number of major crises.
- Potential data-vendor/library differences vs original implementation.
- Risk of overfitting narrative to a small set of historical events.

What this replication supports

- End-to-end pipeline from paper is reproducible in modern Python.
- Topological landscape norms do capture stress periods, especially Lehman.
- TDA signals are useful as *complements* to VIX and classical diagnostics.

What remains open

- Robustness across assets, frequencies, and crisis definitions.
- Better statistical validation and out-of-sample warning evaluation.
- Translation into implementable risk-monitoring tools.

Thank you.

Questions?